

IIT Foundation Module for Class VII

Chemistry

- Theme 1: Matter
 1. Main postulates of kinetic molecular theory of matter.
 2. Explanation of change of state of the matter on the basis of inter-particle space and inter-particle attraction and collision.
 3. Law of conservation of mass
 - Statement and explanation with examples.
- Theme 2: Physical and Chemical Changes
 1. Revise and review the topic on Physical and Chemical Changes taught in earlier classes.
 2. Physical and Chemical changes – Classification with examples.
- Theme 3: Elements, Compounds and Mixtures
 1. Revision of Elements, Compounds and Mixtures taught in earlier classes.
 2. Elements, compounds and mixture – a brief explanation.
 3. Separation of the components of a mixture
 - Emphasis on the principle of separation.
- Theme 4: Atomic Structure
 1. Fundamental subatomic particles present in an atom: electrons, protons, neutrons.
 2. Nucleus and extranuclear parts.
 3. Atomic number and mass number.
- Theme 5: Language of Chemistry
 1. Symbols of elements.
 2. Formulae of compounds.
 3. Chemical equations

- From word equations.
- Law of conservation of mass.
- Balancing simple equations.
- Relate the law to the balancing of simple equations.
- Information gathered from a chemical equation.
- Limitations of a chemical equation:
 - Catalyst, conditions for the reaction, state of reactants and products, nature of the chemical reaction are not gathered from the equation.

- Theme 6: Chemical Reactions

1. Types of reactions:

- Combination
- Decomposition
- Displacement
- Double displacement

2. Reactivity series:

- Metals arranged in order of their reactivity.
- Metal that displaces metal ion from solution is more reactive.
- Predict the reactivity of metals.

3. Endothermic and exothermic processes/reactions.

4. Neutralization reaction.

5. Decomposition reactions to form Oxides.

6. Classification of oxides:

- Acidic
- Basic
- Amphoteric
- Neutral
- Metal oxides are basic; nonmetal oxides are acidic in nature.
- Acidic oxides react with base and basic oxides react with acids. Some oxides, such as ZnO, PbO, react with both acids and bases. These are amphoteric oxides.

- Theme 7: Hydrogen

1. Preparation of hydrogen from water – electrolysis
 - Introduction to terms electrode, electrolyte, electrolysis – detailed process not required.
2. Preparation of hydrogen in the laboratory.
3. Preference of zinc as the metal to be used (with reasons).
4. Choice of dilute acids (other than dil. nitric acid).
5. Bosch's process.
6. Properties and uses of hydrogen.
7. Oxidation and Reduction.

- Theme 8: Water

1. Dissolution of salts in water – meaning and explanation.
2. Universal solvent – meaning.
3. Solutions, suspensions, colloids.
 - Differentiate unsaturated/saturated and supersaturated solutions.
 - Suspensions and colloids.
4. Water of crystallisation.
5. Hydrated and Anhydrous substances, hygroscopic.
6. Reactivity of metals with cold water, hot water and steam (with products formed).
7. Hard and soft water and methods of softening of hard water.
8. Disadvantage of using hard water.
9. Removing hardness of water by boiling or by treating with washing soda.

- Theme 9: Carbon and its Compounds

1. Allotropes of Carbon – definition and explanation.
2. Crystalline and amorphous nature of allotropes of carbon.
3. Uses of diamond, graphite, coke, coal, soot.
4. Laboratory preparation, properties and uses of carbon dioxide
 - Physical properties of Carbon dioxide.
 - Chemical properties of Carbon Dioxide.
 - Acidic nature.
 - Reaction with lime water.
5. Properties and uses of Carbon monoxide

- Emphasis on use as reducing agent in the extraction of iron.
- Emphasize the harmful properties of Carbon monoxide when inhaled – Asphyxia.